INCH-POUND

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DETAIL SPECIFICATION

CONVERTER, LIQUID OXYGEN, 5 LITER, MB-5A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of aircraft converter MB-5A, used for storing and pressurizing 5 liters of liquid oxygen.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index for Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical

DEPARTMENT OF DEFENSE

MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-V-9050	Valves, Oxygen, Pressure Relief, Aircraft
MIL-F-18264	Finishes: Organic, Weapon System, Application and Control of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: OC-ALC/TICLA, 3001 Staff Drive, Ste 1AE1/101A, Tinker AFB, OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1660

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MIL-C-25516	Connectors, Electrical, Miniature, Environment Resistant Type, General Specification for
MIL-L-25567	Leak Detection Compound, Oxygen Systems
MIL-V-25961	Valve, Fill-Buildup-Vent, Liquid Oxygen Converter CRU-50/A
MIL-O-27210	Oxygen, Aviators Breathing, Liquid and Gas
MIL-T-27730	Tape, Antiseize, Tetrafluoroethylene, With Dispenser

STANDARDS

FEDERAL

FED-STD-595	Color Used in Government Procurement

DEPARTMENT OF DEFENSE

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-461	Control of Electromagnetic Interference Emissions and Susceptibility, Requirements for the
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-889	Dissimilar Metals
MIL-STD-1359	Cleaning Methods and Procedures for Breathing Oxygen Equipment
MS33583	Tubing End Double Flare, Standard Dimensions for
MS33584	Tubing End Standard Dimensions for Flared

(Unless otherwise indicated, copies of federal and military specifications, and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robins Ave, Philadelphia, PA 19111-5094)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

NHB 8060.1	Office of Safety and Mission Quality Flammability, Odor, Offgassing, and Compatibility Requirements and Test Procedures for Materials in Environments that Support Combustion

(Application for copies should be addressed to the National Aeronautics and Space and Administration (NASA), NASA Headquarters, Technical Standards Division, Code AE, Washington, DC 20546)

2.3 Non-Government publication. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117	Standard Practice for Operating Salt Spray (Fog) Testing Apparatus
ASTM D2512	Compatibility of Materials with Liquid Oxygen (Impact Sensitivity Threshold and Pass-Fail Technique), Test Method for

(Application for copies should be addressed to the American Society For Testing Materials, 1916 Race St, Philadelphia, PA 19103-1187)

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2.4 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein the text of this documents takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Qualification. The converters furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.4).

3.2 Design and construction. The design and construction shall be in accordance with Figures 1, 2, 3, and 4. The converter shall include provisions for filling with liquid oxygen when either installed or not installed in the aircraft, and for being transported in the filled condition. The design of the components and associated hardware shall be such that the assembly/disassembly of the converter can be accomplished without the use of any specialized tools. All components and hardware shall be designed, protected, and positioned to prevent chafing due to normal service, shipping, and handling environments. Tube assemblies and warming coils shall be a minimum of 0.0625 of an inch (1.58 mm) from the container outer shell where practicable.

3.2.1 Major components. The major components of the converter shall be as follows:

a. Liquid oxygen container (see 3.3.1)

b. Container outer shell (3.3.2)

c. Capacity sensing element (3.3.3)

d. Electrical connectors (3.3.3.1)

e. Combination fill-buildup-vent valve (3.3.4)

f. Relief valve (3.3.5)

g. Pressure control valve (3.3.6)

h. Test port (3.3.7)

i. Overpressure safety device (3.3.8)

3.2.2 Materials. Materials that are not covered by specifications shall be suitable for use in liquid oxygen systems. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the function of the end item.

3.2.2.1 Metals. Metals shall be corrosion-resistant steel or nonferrous metal that has been treated to resist corrosion. Metallic materials that are subject to corrosion in oxygen, salt air, or any other atmospheric conditions likely to occur during the usage of the converter shall be protected against such corrosion in a manner that will not in any way prevent compliance with the performance requirements of this specification.

3.2.2.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.2.3 Nonmetallic materials. Nonmetallic materials shall have the physical properties required for the converter to function without being adversely affected, while exposed to cryogenic temperatures. The contractor shall be responsible for the evaluation of materials for their suitability in each application. The contractor may evaluate materials in accordance with either ASTM D2512 or NASA NHB 8060.1.

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3.2.2.4 Exterior coating. After the cleaning specified (see 3.4) has been completed, the assembled liquid oxygen container and any parts of the other components that are to be painted shall be coated with a primer compatible with the finish to be applied. One or more coats of durable, high quality paint shall be applied as required to achieve a smooth protective finish free of cracks, checks, or runs. The color of the topcoat shall conform to 14187 of FED-STD-595. Organic finishes shall be applied in accordance with MIL-F-18264.

3.2.2.5 Protective treatment. When materials are used in the construction of the converter that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip or scale during normal service life or under extremes of environmental conditions shall not be used.

3.2.3 Tubing and fittings. All openings of the finished converter shall be capped or plugged to prevent thread damage and to prevent the entry of moisture and foreign matter. Aluminum tubing that interconnects components shall be double flared to conform to MS33583. Tube fittings and pipe thread bosses for mating the converter to aircraft lines shall be corrosion-resistant steel. Corrosion-resistant steel tubing shall be single flared to conform to MS33584.

3.2.4 Antiseize tape. Antiseize tape shall be used on all male pipe thread fittings. Antiseize tape shall conform to and shall be applied as specified in MIL-T-27730. Tape shall not be used on flare tube fittings, straight threads, coupling sleeves, or on the outer side of tube flares. None of the tape shall be allowed to enter the inside of a fitting.

3.2.5 Check valves and flow restrictors. Check valves and flow restrictors may be used in the circuit as required for function. Check valves and flow restrictors shall be spring loaded as necessary to avoid any possibility of entrapment of liquid oxygen in a dead end circuit.

3.2.6 Shock mounts. Shock mounts (if provided) and sway clearance due to shock mounting shall be within the specified envelope.

3.2.7 Standard parts. Standard component parts usually used in the aerospace industry shall be encouraged, provided they suit the intended purpose.

3.2.8 Threads. Screw threads shall be in accordance with MIL-S-7742. Pipe threads shall be in accordance with MIL-P-7105.

3.2.9 Dimensions. The dimensions of the converter shall be as specified in figures 1, 2, 3, 4.

3.3 Details of components.

3.3.1 Liquid oxygen container. The liquid oxygen container shall be designed to contain a minimum of 5 liters of liquid oxygen measured at 1 atmosphere (101.3 kPa). The container shall be designed such that in case of contamination the container can be easily and completely cleaned by Overhaul and Repair servicing personnel and then returned to service status. If the container employs a vacuumed space as insulation, it shall be provided with a functional adsorbent. The container shall be baffled to prevent direct exit of liquid oxygen through the vent line during initial stages of the filling operation.

3.3.2 Container outer shell. The outer shell of the container shall be provided with a rupture point having not less than a 0.375 inch (9.525 mm) diameter escape area. Pressure required to rupture the escape area shall be at least 20 percent less than the burst pressure of the outer container shell. The rupture point shall be located in such a position that in case of rupture the danger to personnel will be at a minimum.

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3.3.3 Capacity sensing element. The capacity sensing element shall consist of an electrical capacitance having characteristics that provide for a linear relationship between electrical capacitance and volume sensed. The element shall be designed for use with indicator system components to indicate the amount of liquid oxygen in the liquid oxygen container and shall be an integral part of the container, but may be either removable or permanently installed. The element shall not be electrically grounded to the container. The interference limits shall be in accordance with MIL-STD-461 and MIL-STD-462.

3.3.3.1 Electrical connectors. The high impedance sensing element lead shall be terminated in a connector conforming to MIL-C-25516, B polarity. The low impedance lead shall be terminated in a connector conforming to MIL-C-25516, E polarity. The connectors shall provide waterproof connections. The connectors shall be provided with dust caps and tethered to the converter. Storage clips shall be provided for these caps near the electrical connectors.

3.3.4 Combination fill-buildup-vent valve. The combination fill-buildup-vent valve, as a component only, shall meet the performance requirements of MIL-V-25961.

3.3.5 Relief valve. The relief valve, as a component only, shall meet the performance requirements of MIL-V-9050, Type II.

3.3.6 Pressure control valve. The pressure control valve to be used shall be a pressure closing valve.

3.3.7 Test port. A test port with a cap and tether shall be incorporated in the tubing and located in such a manner that the entire converter assembly can be pressurized through the test port.

3.3.8 Overpressure safety device. A frangible safety disc or other secondary overpressure device shall be incorporated in the plumbing of the converter at a location to insure rapid dumping of the head pressure from the converter in the event the pressure relief valve fails to open. The overpressure safety device shall be designed to remain in the open position if actuated.

3.4 Cleaning.

3.4.1 Preassembly of converter. Prior to the assembly of the converter, the internal and external surfaces of the components shall be degreased in accordance with MIL-STD-1359 or qualifying activity approval of the contractor's procedure.

3.4.2 After welding and brazing. After all welding and brazing operations have been completed, all weld areas shall be cleaned, and weld spatter shall be removed by suitable mechanical or abrasive cleaning techniques. All flux residue shall be thoroughly removed in order to preclude corrosive effects which may occur undetected beneath the paint and result in early failure of the converter in service. The cleaning solutions and the processes that are used shall be compatible with the converter and its expected service life.

3.4.3 Purging. The completed converter, prior to preparation for delivery, shall be purged with hot, dry oxygen, conforming to MIL-O-27210, Type I or nitrogen conforming to BB-N-411, Type I, Class 1, Grade B, until all traces of the contaminants are removed. The temperature of the purging gas shall not exceed 250 F (121 C) at the converter inlet. The purge pressure shall be 50 $\pm$ 5 psig (345 $\pm$ 34 kPa). After purging, charge the container with gaseous oxygen at 25 $\pm$ 5 psig (172 $\pm$ 34 kPa) and close all openings with approved closures.

3.4.4 Odors. The interior of the converter shall be substantially free from odor. The converter shall be free of toxic gases or by-products.

3.5 Identification of product. The converter shall be marked for identification in accordance with MIL-STD-130.

3.5.1 Vacuum performance plate. A metal plate shall be attached to the converter to provide spaces and columns to show results and dates of evaporation loss tests. The plate shall provide sufficient space for

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permanently recording results of at least 12 tests. Results and dates of all the evaporation loss tests conducted by the converter manufacturer shall be lightly but legibly recorded upon the plate by the manufacturer.

3.5.2 Warning notes. A warning note shall be located in a conspicuous place on the container. The note may be applied by decal, plate, or stencil. The following information, in .25 inch (6.35 mm) high red letters on a white or frosted aluminum background, shall be provided:

CAUTION
HIGH VACUUM CONTAINER
HANDLE WITH CARE

3.6 Workmanship. The converters shall be uniform in quality and shall be free from irregularities, defects, or foreign material which could adversely affect safety, performance, reliability, or durability.

3.7 Performance.

3.7.1 Final assembly pressure. When the converter is subjected to the final assembly pressure of 95 psig (655kPa) as specified (see 4.6.1), it shall not leak except for the leakage allowed for installed components by their respective specifications.

3.7.2 Filling time. The time required to fill the converter, which shall have been initially at room temperature, when tested as specified (see 4.6.2), shall not exceed 5 minutes.

3.7.3 Evaporation loss.

3.7.3.1 Initial evaporation loss in vented condition. The converter, when tested as specified (see 4.6.3.2), shall not have a weight loss rate greater than 2.5 pounds (1.13 kg) for a 24-hour period with the unit vented to the atmosphere.

3.7.3.2 Second evaporation loss in vented condition. The converter, when tested as specified (see 4.6.3.3), shall meet the requirements specified (see 3.7.3.1).

3.7.3.3 Evaporation loss in pressurized condition. The converter, when tested as specified (see 4.6.3.4), shall not have a weight loss rate greater than 3.0 pounds (1.36kg) for a 24 hour period. If this amount is exceeded then 6.0 pounds (2.72 kg) for a 48-hour period may be substituted.

3.7.3.4 Buildup time. The time required to attain the buildup pressure of 70 psig (483 kPa), when tested as specified (see 4.6.4), shall not exceed 5 minutes.

3.7.3.5 Dielectric strength. The leakage current of the converter sensing element, when measured as specified (see 4.6.5), shall not exceed 1.0 milliampere.

3.7.3.6 Insulation resistance. The insulation resistance of the converter sensing element, when measured as specified (see 4.6.6), shall be a minimum of 10 megohms between the converter ground and each electrode, and a minimum of 20 megohms between the electrodes.

3.7.4 Sensing element calibration.

3.7.4.1 Empty converter. The capacitance of the sensing element, when inspected as specified (see 4.6.7.1), shall be 63.5 $\pm$1 picofarads.

3.7.4.2 Capacitance. The capacitance of the sensing element, when tested as specified (see 4.6.7.2), shall be between the maximum and minimum limits as derived from the following equations:

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Cmax = (2.33 x W) + 64.0

Cmin = (2.25 x W) + 63.0

C = Capacitance in picofarads (pf)

W = Exact weight of the liquid oxygen in pounds at the test point.

3.7.5 Inner vessel pressure.

3.7.5.1 Inner vessel. When the inner vessel is subjected to 200 ±5 psig as specified (see 4.6.8.1), it shall not show any evidence of material failure.

3.7.5.2 Permanent volumetric expansion. The permanent volumetric expansion of the inner vessel, when tested as specified (see 4.6.8.2), shall not exceed 10 percent of the volumetric expansion at the test pressure. There shall be no evidence of mechanical or material failure.

3.7.5.3 Pressure closing valve leakage. When the converter buildup pressure conditions are applied to the pressure closing valve, the leakage past the valve seat shall not exceed 0.02 liter per minute measured at normal temperature and pressure, with a pressure differential of 5 to 10 psi being maintained across the pressure closing valve seat (see 4.6.8.3).

3.7.6 Delivery rate.

3.7.6.1 Delivery rate. The converter, when tested as specified (see 4.6.9.1), shall function normally and deliver oxygen on demand for at least 45 minutes and until not more than one pound of oxygen remains in the converter.

3.7.7 Overpressure safety.

3.7.7.1 Overpressure safety (standard temperature). The overpressure safety device, when tested as specified (see 4.6.10.2), shall not leak at a pressure below 175 psig (1206.58 kPa), shall open within a pressure range of 200 psig (1378.95 kPa) minimum and 250 psig (1723.68 kPa) maximum, and shall allow a minimum flow of 150 lpm at opening pressure.

3.7.7.2 Overpressure safety (low temperature). The overpressure safety device, when tested as specified (see 4.6.10.3), shall not leak at a pressure below 175 psig (1206.58 kPa), shall open within a pressure range of 200 psig (1378.95 kPa) minimum and 315 psig (2171.84 kPa) maximum, and shall allow a minimum flow of 150 lpm at opening pressure.

3.7.7.3 Overpressure safety (high temperature). The overpressure safety device, when tested as specified (see 4.6.10.4), shall not leak at a pressure below 175 psig (1206.58 kPa), shall open within a pressure range of 175 psig (1206.58 kPa) minimum and 250 psig (1723.68 kPa) maximum, and shall allow a minimum flow of 150 lpm at opening pressure.

3.7.8 Vibration.

3.7.8.1 Vibration. The converter, when tested as specified (see 4.6.11.1), shall show no evidence of mechanical or material failure and the evaporation loss shall not exceed 2.5 pounds (1.13 kg).

3.7.8.2 Acceleration. The converter, when tested as specified (see 4.6.12), shall show no evidence of mechanical or material failure and the evaporation loss shall not exceed 2.5 pounds (1.13 kg).

3.7.9 Shock.

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3.7.9.1 Shock (procedure I). The converter, when tested as specified (see 4.6.13.1 and 4.6.13.4) shall show no evidence of mechanical or material failure and the evaporation loss shall not exceed 2.5 pounds (1.13 kg).

3.7.9.2 Shock (procedure II). The converter, when tested as specified (see 4.6.13.1 and 4.6.13.5), shall not break free from the mounting provisions.

3.7.10 Salt fog. The converter, when tested as specified (see 4.6.14), shall show no evidence of corrosion or mechanical failure.

3.7.11 Gunfire. The converter, when tested as specified (see 4.6.15), shall show no evidence of shattering. The installation mounting base of the converter shall withstand the forces exerted by the gunfire test.

3.7.12 Reliability. The converter, when tested as specified (see 4.6.16), shall have a specified mean time between failures (MTBF) of not less than 300 hours at the 90 percent lower confidence limit. (This is equivalent to a minimum acceptable MTBF of 200 hours, and to a reliability of 0.9933 for a 2-hour mission.)

3.7.13 Maintainability. The total corrective maintenance time, when performed as specified (see 4.6.17), shall not exceed 1 hour of maintenance downtime.

3.7.14 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.7.15 Weight. The empty weight of the complete converter, including shock mounts if provided and the receptacle halves of the quick disconnect couplings, shall not be greater than 15 pounds (6.29 kg).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 Qualification inspection. Qualification inspection shall consist of all the examinations and tests of this specification.

4.2.1 Qualification samples. Qualification samples shall consist of the following:

a. Two each complete converter assemblies.

b. One each inner vessel complete with all lines and fittings.

c. One each sensing element.

d. One each combination fill-buildup-vent valve.

e. One each relief valve.

f. One each pressure control valve.

g. Three each overpressure safety devices.

h. A Statement of Certification (with supporting test data) from the manufacturer that the converters conform to the reliability requirements cited (see 3.7.12 and 4.6.16).

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4.2.2 Qualification sample for the qualifying activity. Along with the qualification test report, the manufacturer shall submit one converter to the qualifying activity. This sample may be a tested sample, except that it shall not be subjected to the gunfire test.

4.3 Conformance inspection. The conformance inspection shall consist of the following:

a. Individual inspection (see 4.3.1)

b. Sampling inspection (see 4.3.2)

4.3.1 Individual inspection

4.3.1.1 Converters. Each converter shall be subjected to the following methods of inspection:

a. Visual examination (see 4.5.1)

b. Inner vessel pressure (see 4.6.8)

c. Filling time (see 4.6.2)

d. Buildup time (see 4.6.4)

e. Dielectric strength (see 4.6.5)

f. Insulation resistance (see 4.6.6)

g. Sensing element calibration (empty converter) (see 4.6.7.1)

h. Final assembly pressure (see 4.6.1)

I. Evaporation loss (see 4.6.3)

j. Overpressure safety (see 4.6.10)

k. Delivery rate (room temperature)(see 4.6.9.2) the delivery rate shall be monitored for 20 minutes and then the test shall be terminated.

4.3.2 Sampling inspection. The lot size for sampling plan inspection shall be 50 units (converters). One out of every 50 units that have been manufactured under essentially the same conditions using the same facilities and equipment shall be selected for sampling inspection. Continuous production is not required. Component parts and material need not be delivered from the same batch or lot. The following tests or inspections are required:

a. Dimensions (see 4.5.1.1)

b. Delivery rate (low/high temperature) (see 4.6.9.3 and 4.6.9.4)

c. Sensing element calibration (filled converter) (see 4.6.7.2)

d. Overpressure safety (standard temperature) (see 4.6.10.2)

e. Delivery rate (room temperature) (see 4.6.9.2)

4.3.2.1 Rejection and retest. If one or more items selected from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative and notify the procuring activity in writing the

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cause of failure, the action taken to preclude recurrence, and the impact this failure might have on scheduled deliveries. After correction, all of the sampling inspection shall be repeated.

4.3.2.2 Continuation of inspection. For production reasons, individual inspection or other sampling plans may be continued pending the investigation of a sampling inspection failure. Final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of this specification.

4.3.3 Defects in items already accepted. The investigation of a test failure could indicate that defects might exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.4 Inspection conditions.

4.4.1 Oxygen. Unless otherwise specified, the oxygen used in testing the converters shall conform to MIL-O-27210, Type I or Type II.

4.4.2 Temperature and pressure. Unless otherwise specified, tests shall be conducted at local ambient temperatures and barometric pressure. Corrections shall be made to provide agreement with the temperature and pressure calibration of the instruments. Inspection data provided by any instrument not calibrated to normal temperature and pressure (NTP) conditions shall be corrected to determine NTP requirements. NTP conditions are 29.92 inches of mercury (101.3 kPa) and 70°F (21.1°C).

4.5 Inspection methods.

4.5.1 Visual examination. Every converter shall be visually examined to determine conformance to the requirements of this specification. The classification of defects shall be according to table I, unless otherwise specified. All defects not corrected shall be cause for rejection. Minor color shade deviation need not be corrected.

TABLE I. Classification of defects for visual examination of the converter.

Critical
1. Material imperfections - foreign matter embedded
2. Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws.
3. Any component missing, malformed, fractured, or otherwise damaged.
4. Any component loose or otherwise not securely retained.
5. Incorrect assembling or improper positioning of components.
6. Any functioning part that works with difficulty.
7. Faulty workmanship or other irregularities.

4.5.1.1 Dimensions. The converter shall be checked dimensionally to determine conformance to the dimensions specified (see 3.2.9).

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4.6 Performance.

4.6.1 Final assembly pressure. The converter, including all its fittings and connections, shall be subjected to the pneumatic pressure that is specified as the final assembly pressure (see 3.7.1). While this pressure is maintained, the converter shall be tested by applying leak detection compound that conforms to MIL-L-25567, type I, to all fittings and connections of the converter and then examining all fittings and connections of the converter for leakage. The pressure shall then be increased to a level sufficient to open the pressure relief valve and to demonstrate that the required flow capacity can be achieved without excessive pressure. If there is any leakage from the converter, except for the leakage's that are allowed for installed components by their respective specifications, the converter shall be rejected (see 3.7.1). Nitrogen, per BB-N-411, type I, grade B, class 1 may be used in lieu of oxygen for the inspection.

4.6.2 Filling time. The time required to fill the converter to its specified capacity by transfer of liquid oxygen using standard transfer equipment shall be determined. The filling pressure at the converter filler valve shall be maintained at 30 psig (207 kPa). The time required to fill the converter shall be within the time specified (see 3.7.2).

4.6.3 Evaporation loss.

4.6.3.1 Evaporation loss tests. The evaporation loss tests in the vented condition may be conducted on the container only. The weight losses shall be recorded on the vacuum performance test plate as specified (see 3.5.1).

4.6.3.2 Initial evaporation loss in vented condition. The container shall be filled with liquid oxygen to rated capacity and allowed to stand in vented condition for a minimum period of two hours for equilibrium to be established. After which, with the unit vented to the atmosphere, the rate of loss in weight for a 24-hour period shall be recorded. The container shall pass the requirements specified (see 3.7.3.1).

4.6.3.3 Second evaporation loss in vented condition. After a lapse of at least 30 days from the above test, the vented evaporation loss test shall be conducted in the same manner and the evaporation loss for a 24-hour period recorded. The container shall meet the requirements specified (see 3.7.3.1).

4.6.3.4 Evaporation loss in pressurized condition. Upon completion of the vented evaporation loss tests, the complete converter assembly including all components shall be subjected to a 48-hour evaporation loss test in the pressurized condition. The converter shall be placed in the pressurized condition and the initial weight reading shall be taken immediately after the converter has been filled. The loss in weight for a 48-hour period shall be recorded. The converter assembly shall pass the requirements specified (see 3.7.3.3).

4.6.4 Buildup time. The time required to attain buildup pressure shall be determined with a converter that has been empty for a minimum of 30 minutes. The converter shall be filled with liquid oxygen to design capacity and immediately after filling, the converter shall be placed in the pressurized (buildup) condition. The time required to attain buildup pressure shall be within the time specified (see 3.7.3.4).

4.6.5 Dielectric strength. The sensing element shall be installed in an empty converter and the dielectric strength of the sensing element shall be measured at a voltage of 500 volts, rms, at a commercial frequency. The voltage shall be maintained for a period of 10 seconds between the converter ground and each lead wires, and between the two lead wires. The sensing element of the converter shall pass the requirements specified (see 3.7.3.5).

4.6.6 Insulation resistance. The sensing element shall be installed in an empty converter and the insulation resistance of the sensing element shall be measured at 50 volts, direct current, between each electrode and the converter ground, and between the electrodes. The sensing element of the converter shall pass the requirements specified (see 3.7.3.6).

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4.6.7 Sensing element calibration.

4.6.7.1 Empty converter. The sensing element shall be installed in an empty converter and the capacitance of the sensing element measured. The capacitance shall be within the limits specified (see 3.7.4.1).

4.6.7.2 Capacitance. The converter assembly shall be filled to rated capacity with liquid oxygen. The liquid oxygen shall then be removed in 4 increments of 2.5 ± 0.2 pounds (1.13 ± 0.09 kg). The capacitance of the sensing element shall be measured with the converter filled and at each increment. The capacitance shall be within the limits specified (see 3.7.4.2).

4.6.8 Inner vessel pressure inspection.

4.6.8.1 Inner vessel. The test shall be conducted before the outer vessel is added. The inner vessel shall be charged to the pressure specified (see 3.7.5.1) and maintained for at least one minute. The inner vessel shall meet the requirements specified (see 3.7.5.1).

4.6.8.2 Permanent volumetric expansion. This test shall be conducted before the outer vessel is added. The inner vessel, complete with all lines and fittings, shall be subjected to a hydrostatic or pneumatic test in a water jacket in which the vessel is totally submerged and free to expand in all directions. The difference in temperature between the water in the jacket and the water or gas in the vessel shall not exceed 5°F (2.8°C). The test pressure shall be at least 200 psig (1379 kPa) and shall be maintained for at least one minute or sufficiently longer to insure complete expansion of the vessel. Any internal pressure previously applied shall not exceed 90 percent of the test pressure. The pressure gage shall permit readings to an accuracy of one percent. The expansion gage shall permit readings to an accuracy of one percent of 0.1 cc. The expansion shall be recorded in cc. The water in the expansion gage shall come to rest quickly. If the water in the expansion gage drops below its original position or does not come to rest quickly after the pressure is released, the test shall be repeated. The inner vessel shall pass the requirements specified (see 3.7.5.2).

4.6.8.3 Pressure closing valve leakage. A pneumatic pressure source shall be applied to the inlet of the pressure closing valve. Suitable pressure gages and control valves shall be installed in the inlet and outlet of the pressure closing valve. Pressure shall be applied gradually to the inlet of the pressure closing valve until the pressure closing valve closes. The inlet pressure shall be gradually increased until it exceeds the closing pressure by 5 to 10 psig. The allowed leakage shall not exceed the limits specified (see 3.7.5.3). Nitrogen conforming to BB-N-411, type I, class 1, grade B may be used for this inspection.

4.6.9 Delivery rate.

4.6.9.1 Delivery rate. The evaporation tubing shall consist of 100 ± 5 feet (30.48 ± 1.52 m) of 5/16 inch (7.94 mm) outside diameter aluminum tubing. The supply port of the converter shall be connected to a breathing machine through the evaporation tubing. The breathing machine shall be set up to simulate the complete breathing characteristics of three human subjects, breathing in the flow patterns shown in figure 5. The converter shall be filled to rated capacity prior to the test. After 3 minutes of stabilization, the converter pressure at the supply port shall remain between 55 and 90 psig (379 and 620 kPa) throughout the test. A pressure drop below 55 psig (379 kPa) during the first three minutes of the test is acceptable.

4.6.9.2 Delivery rate (room temperature). The converter assembly shall be conditioned at room temperature for 3 hours. After conditioning and while still at this temperature, the converter shall be tested as specified (see 4.6.9.1). The converter shall pass the requirements specified (see 3.7.6.1).

4.6.9.3 Delivery rate (low temperature). The converter assembly shall be conditioned at a temperature of -65 ± 2°F (-53.9 ± 1.1°C) for 3 hours. After conditioning and while still at this temperature, the converter shall be tested as specified (see 4.6.9.1). The converter shall pass the requirements specified (see 3.7.6.1).

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4.6.9.4 Delivery rate (high temperature). The converter assembly shall be conditioned at a temperature of 260 ± 2°F (126.7 ± 1.1°C) for 3 hours. After conditioning and while still at this temperature, the converter shall be tested as specified (see 4.6.9.1). The converter shall pass the requirements specified (see 3.7.6.1).

4.6.10 Overpressure safety.

4.6.10.1 Overpressure safety. The overpressure safety test shall be performed on the overpressure safety device prior to installation on the converter. The inlet of the overpressure safety device shall be subjected to a gradually increasing pressure until opening occurs.

4.6.10.2 Overpressure safety (standard temperature). The overpressure safety device shall be conditioned at 72 ± 5 F (22 ± 3 C) for three hours. After conditioning, and while still at this temperature, the safety device shall be tested as specified (see 4.6.10.1). The safety device shall pass the requirements specified (see 3.7.7.1).

4.6.10.3 Overpressure safety (low temperature). The overpressure safety device shall be conditioned at -200 ± 5°F (-129 ± 3°C) for three hours. After conditioning, and while still at this temperature, the safety device shall be tested as specified (see 4. 6.10.1). The safety device shall pass the requirements specified (see 3.7.7.2).

4.6.10.4 Overpressure safety (high temperature). The overpressure safety device shall be conditioned at 260 ± 5°F (127 ± 3°C) for three hours. After conditioning, and while still at this temperature, the safety device shall be tested as specified (see 4.6.10.1). The safety device shall pass the requirements specified (see 3.7.7.3).

4.6.11 Vibration.

4.6.11.1 Vibration. The converter shall be attached by its mounting plate to a rigid test fixture capable of transmitting the vibration conditions specified herein. The vibration test shall be conducted at room temperature with the converter filled with liquid Oxygen and in the buildup condition. Tests shall be conducted under both resonance and cycling conditions as specified (see 4.6.11.2 and 4.6.11.3). Vibration displacement, velocity or acceleration shall be measured and interpreted at the specific instantaneous test frequency applied. The vibration test table shall actually move through the displacement corresponding to the required acceleration at all frequencies. Local impacting of loose components which results in high "noise" content of the applied vibration shall not be considered as vibration input. If such "noise" is encountered, a low pass frequency filter which cuts off at approximately twice the test frequency shall be employed to reject the "noise." After the vibration test, the converter shall be filled, if necessary, with liquid oxygen and then vented to the atmosphere for a 24-hour period. The converter shall pass the evaporation loss requirements and the structural requirements specified (see 3.7.3 and 3.7.8.1). The converter shall then be subjected to and pass the delivery rate (room temperature) test specified (see 4.6.9.2).

4.6.11.2 Resonance. Resonant nodes of the converter shall be determined by varying the frequency of applied vibration slowly through the specified range at vibratory accelerations not exceeding those shown in figure 6. Individual resonance surveys shall be conducted with vibration applied along each axis of any set of three mutually perpendicular axes of the converter. The converter shall be vibrated at the indicated resonant conditions for the periods shown in the vibration test schedule, table II, and with the applied double amplitudes of vibratory accelerations in figure 6. These periods of vibration shall be accomplished with vibration applied along each of the three mutually perpendicular axes of vibration. When more than one resonance is encountered with vibration applied along any one axis, each resonance shall be sustained for the period shown in the applicable portion of the vibration test schedule. If more than four resonance's are encountered with vibration applied along any one axis, the four most severe resonance's shall be chosen for test.

TABLE II. Vibration test schedule.

(Times shown refer to one axis of vibration)

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Number of resonance's	0	1	2	3	4
Total vibration time at resonance 1/	-	30 Min	1 hr	1-1/2 hr	2 hr
Cycling time	3 hr	2-1/2 hr	2 hr	1-1/2 hr	1 hr

1/ 30 minutes at each resonance

4.6.11.3 Cycling. The converter shall be vibrated under the cycling conditions specified herein for the applicable periods listed in table II. The frequency shall be cycled between 5 and 500 cps and return in 15-minute cycles at an applied double amplitude of 0.036 inch or an applied acceleration of ± 10 g whichever is the lower "g" value. The rate of change of frequency shall be logarithmic.

4.6.12 Acceleration inspection. The converter, with the probe assembled, shall be filled to rated capacity with liquid oxygen, allowed to reach operating pressure under no flow conditions, and then subjected to an acceleration inspection as specified. Acceleration shall be applied by a centrifuge. The converter shall be subjected to one minute of steady acceleration, in turn, in each direction along each of it's three orthogonal axes. The forward-backward and lateral forces applied shall be 10g. The vertically downward force applied shall be 14g. The vertically upward force applied shall be 6g. After the acceleration inspection, the converter shall be vented to the atmosphere for a 24 hour period. The converter shall pass the evaporation loss requirements and the structural requirements specified (see 3.7.8.2). The converter shall then be subjected to and pass the delivery rate (room temperature) inspection specified (see 4.6.9.2).

4.6.13 Shock.

4.6.13.1 Shock test. The converter, filled to rated capacity with liquid oxygen and with the probe assembled, shall be tested per the following.

4.6.13.2 Accelerometer mounting. The monitoring accelerometer shall be rigidly attached to the test item support fixture at or near the attachment point of the test item.

4.6.13.3 Shock pulse. The shock pulse shall be as shown on figure 7. All points of the acceleration wave form obtained shall lie within the area enclosed by the tolerance limit lines.

4.6.13.4 Procedure I, basic design test. Three shocks in each direction shall be applied along three mutually perpendicular axes of the test item (total of 18 shocks). After the shock test, procedure I, the converter shall be vented to the atmosphere for a 24 hour period. The converter shall pass the evaporation loss and the structural requirements specified (see 3.7.9.1). The converter shall then be subjected to and pass the delivery rate (room temperature) test specified (see 4.5.16.1).

4.6.13.5 Procedure II crash safety test. The converter shall be attached by its normal points of attachments. It shall be subjected to two shocks in each direction along three mutually perpendicular axes of the equipment (total of 12 shocks). The converter shall pass the structural requirements specified (see 3.7.9.2).

4.6.14 Salt fog. The converter, with its ports adequately plugged and the probe assembled, shall be subjected to the salt fog test specified in ASTM B 117. The length of the salt spray test shall not be less than 50 hours. The converter shall pass the requirements specified (see 3.7.10). The converter shall then be subjected to, and pass the following tests:

a. Visual examination. (see 4.5.1)

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b. Delivery rate (room temperature) (see 4.6.9.2).

c. Final assembly pressure test (see 4.6.1).

d. Sensing element calibration (see 4.6.7).

e. Evaporation loss (see 4.6.3.4).

4.6.15 Gunfire test. The converter containing all plumbing and components shall be mounted on a frame capable of withstanding the maximum force which the converter will be subjected to by the gunfire test. The converter shall be filled to rated capacity with liquid oxygen and the nominal operating pressure of the converter maintained. The converter shall be fired upon from a distance of approximately 50 yards (45.7 m) by a .50 caliber armor piercing incendiary bullet which has been tumbled. The bullet shall strike the converter in the liquid phase. There shall be no evidence of shattering, and the converter assembly shall remain in one piece except components in a direct line of fire. Shattering is defined as fragmentation of basic materials. Separation of small components from the converter when they are located in the immediate vicinity of the impact area is satisfactory. The converter shall pass the requirements specified (see 3.7.11).

4.6.16 Reliability. The converter shall be subjected to the delivery rate testing at room temperature (see 4.6.9.2). A failure shall be identified as any malfunction which causes the converter to deliver an insufficient flow of oxygen or to maintain an improper operating pressure. No parts shall be replaced as preventive maintenance during reliability testing. The converter shall be checked at least once every 24 hours during reliability testing to determine if a failure has occurred (see 3.7.12).

4.6.17 Maintainability. Maintainability tests shall be performed to show compliance. All corrective maintenance tasks for expected failure modes shall be conducted by simulating failures. Each maintenance task shall be performed at least three times to determine average values. Only field level maintenance tasks shall be considered. The demonstration shall include the procedure to troubleshoot, fault isolation, removal (based only on the installation provisions of the item), fault correction, adjustment and realignment, final test, and replacement, but shall exclude supply and administrative time. Upon completion, the converter shall be operated to demonstrate normal functioning and will deliver oxygen on demand. The converter shall pass the requirements specified (see 3.7.13).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contracting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The converter covered by this specification is intended to be used in military fighter aircraft liquid oxygen systems for storing and pressurizing liquid oxygen for subsequent conversion to the gaseous phase as required for aircrew breathing during flight.

6..2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification, including any amendments.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Applicable Government part number.

c. Applicable Qualified Products List.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19328 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from
OC-ALC/TICLA, 3001 Staff Drive, OK 73145-5990.

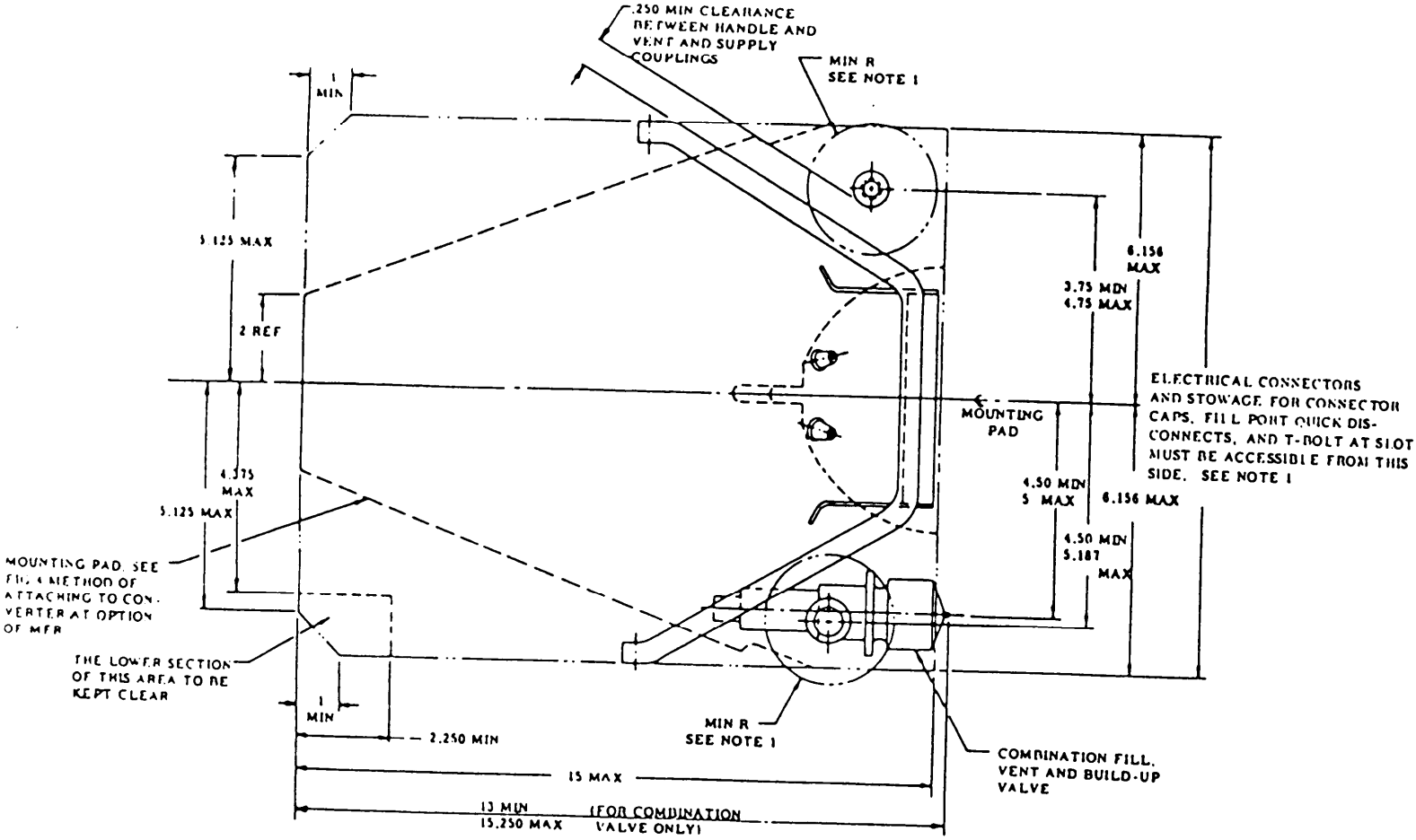
6.4 International standardization. Certain provisions of this specification are the subject of international standardization agreement (ASCC Air Std 14/3). When amendment, revision or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.5 Subject term (key word) listing.

Capacitance
Container
Delivery rate
Evaporation loss
Gunfire
Relief valve
Sensing element
Valve, fill-buildup-vent
Valve, pressure control
30 psi
70 psi

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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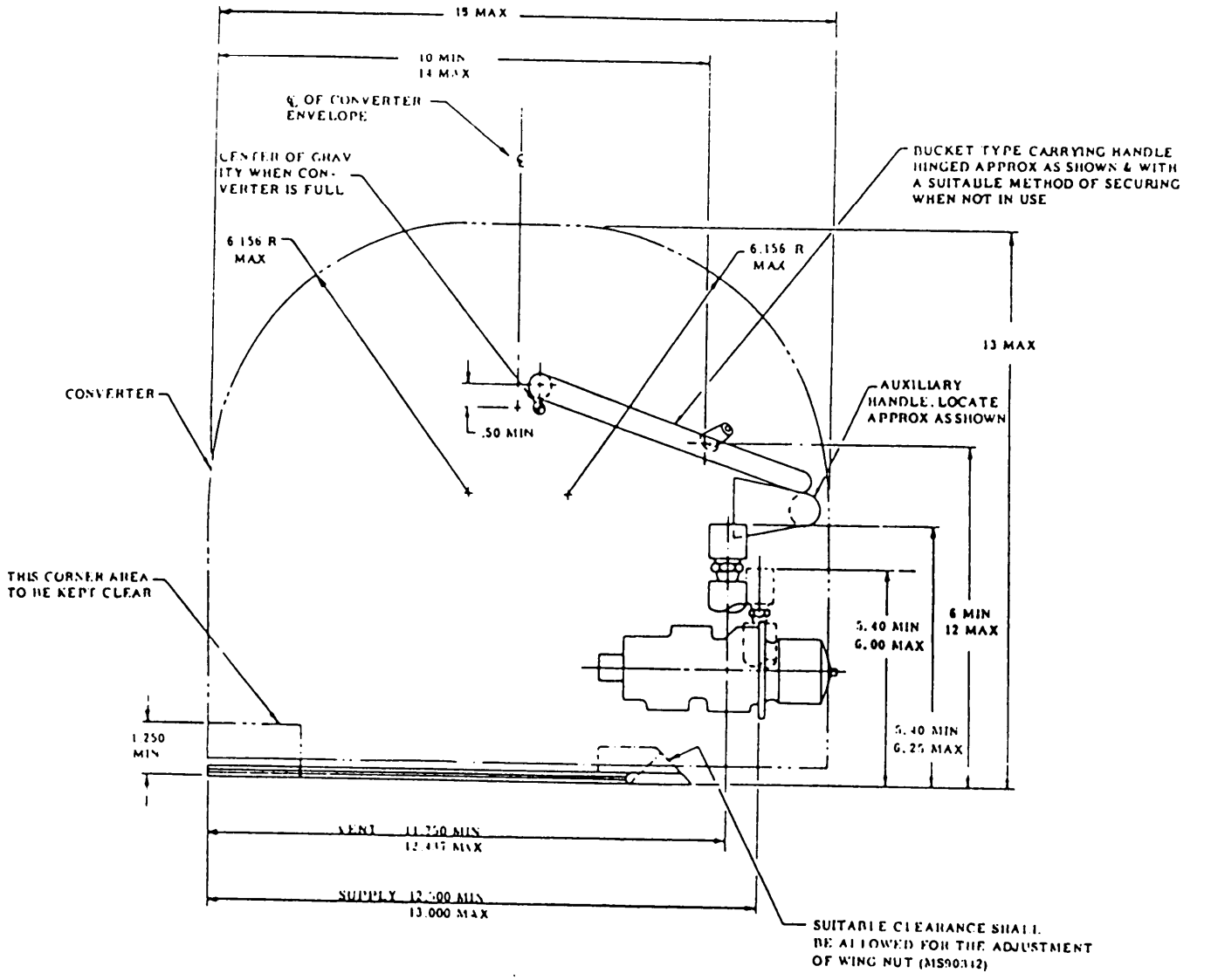
1. 1.5 inch minimum radius from center line of supply and vent ports shall be provided for glove clearance for mating and disconnecting couplings. Radius is not restricted to maximum envelope of converter.

2. All components shall be within the maximum specified envelope.

3. Unless otherwise specified, dimensions in inches.

FIGURE 1. Top view.

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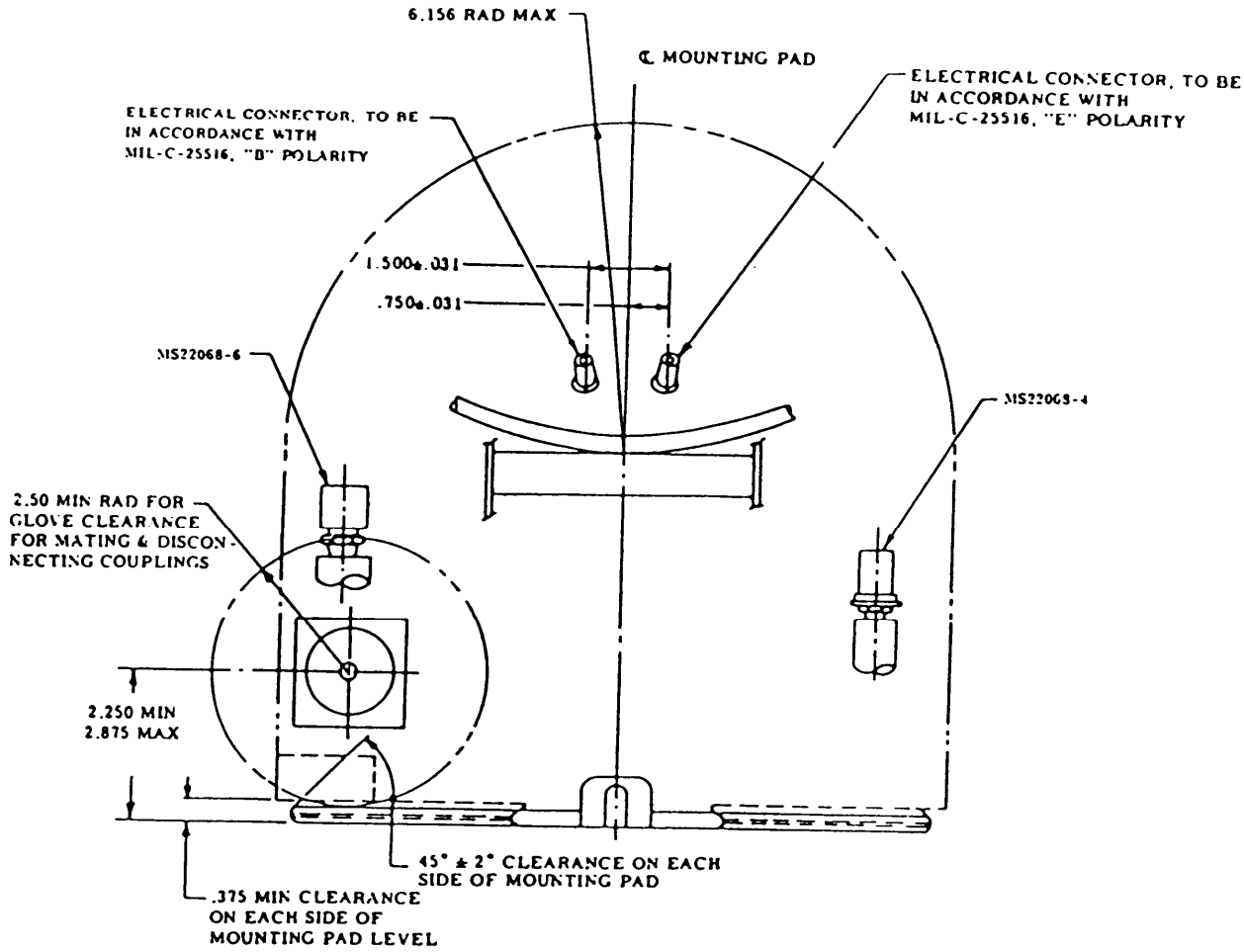
1. 1.5 inch minimum radius from center line of supply and vent ports shall be provided for glove clearance for mating and disconnecting couplings. Radius is not restricted to maximum envelope of converter.

2. All components shall be within the maximum specified envelope.

3. Unless otherwise specified, dimensions in inches.

FIGURE 2. Side view.

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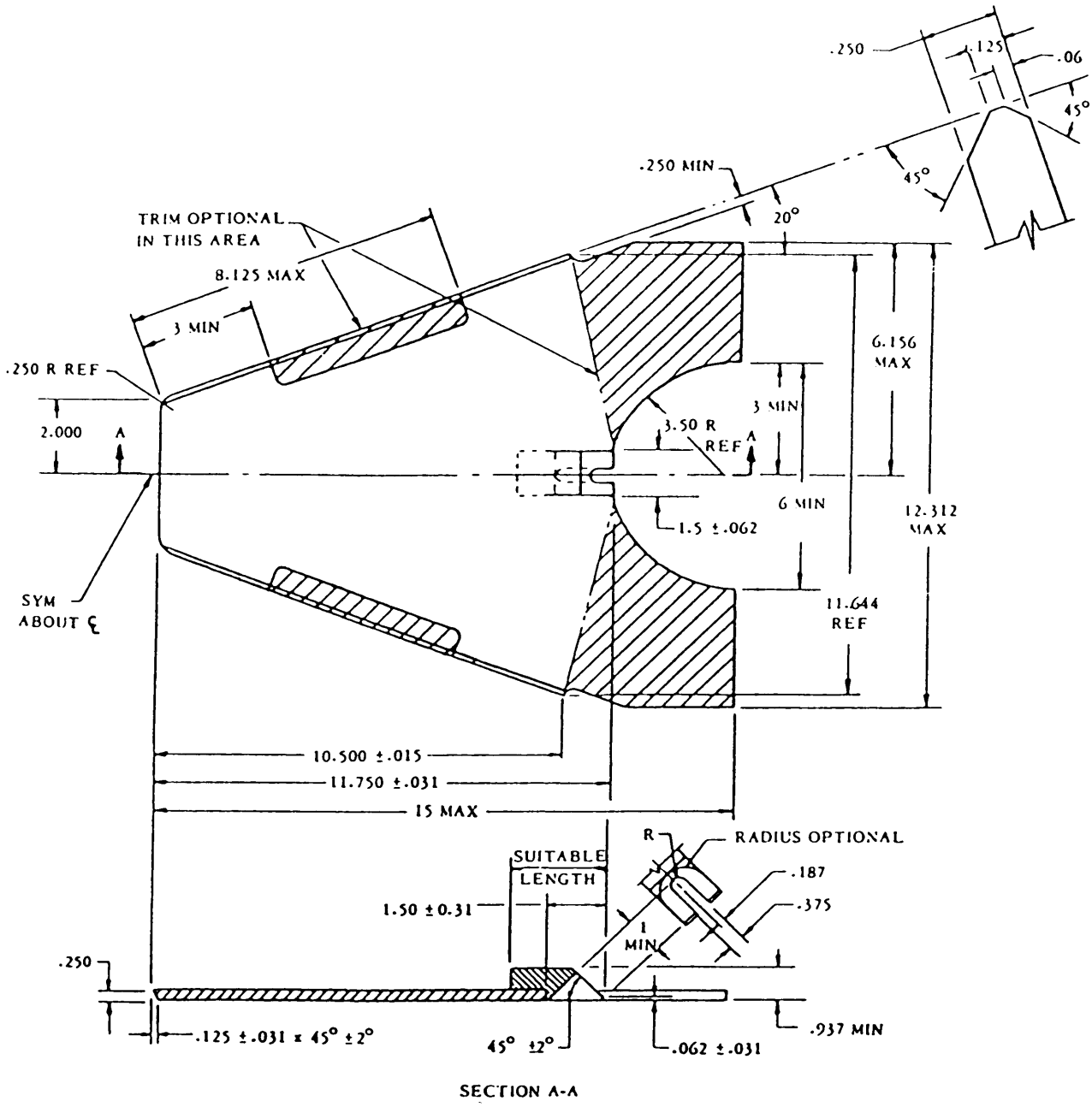
NOTES:

1. All components shall be within the maximum specified envelope.

2. Unless otherwise specified, dimensions in inches.

FIGURE 3. Front view.

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NOTES:

1. This mounting base is designed to mate with the mounting bracket shown on MS90341.

2. Unless otherwise specified, dimensions in inches. Tolerances: Decimals ± .010, angles ±.5°.

FIGURE 4. Mounting base.

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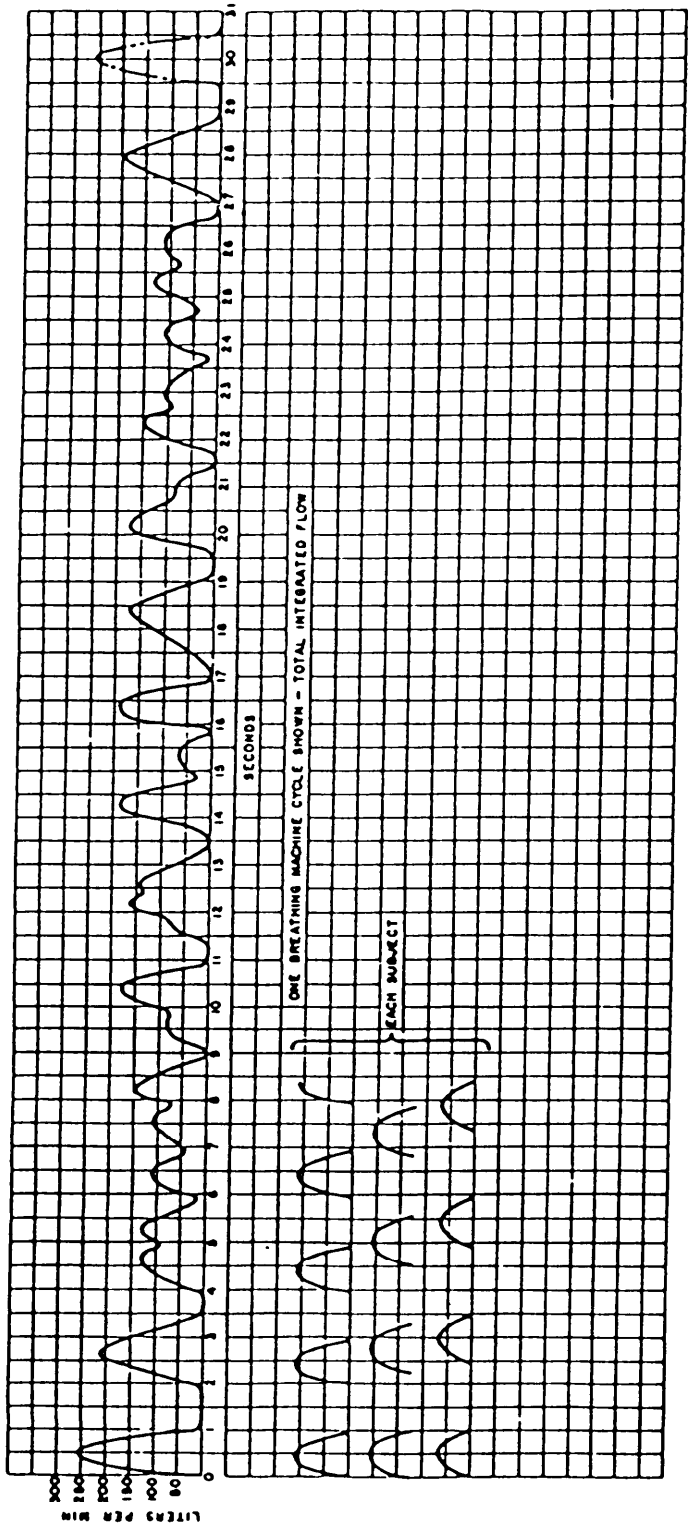


FIGURE 5. Delivery rate flow pattern, three subjects, moderate work.

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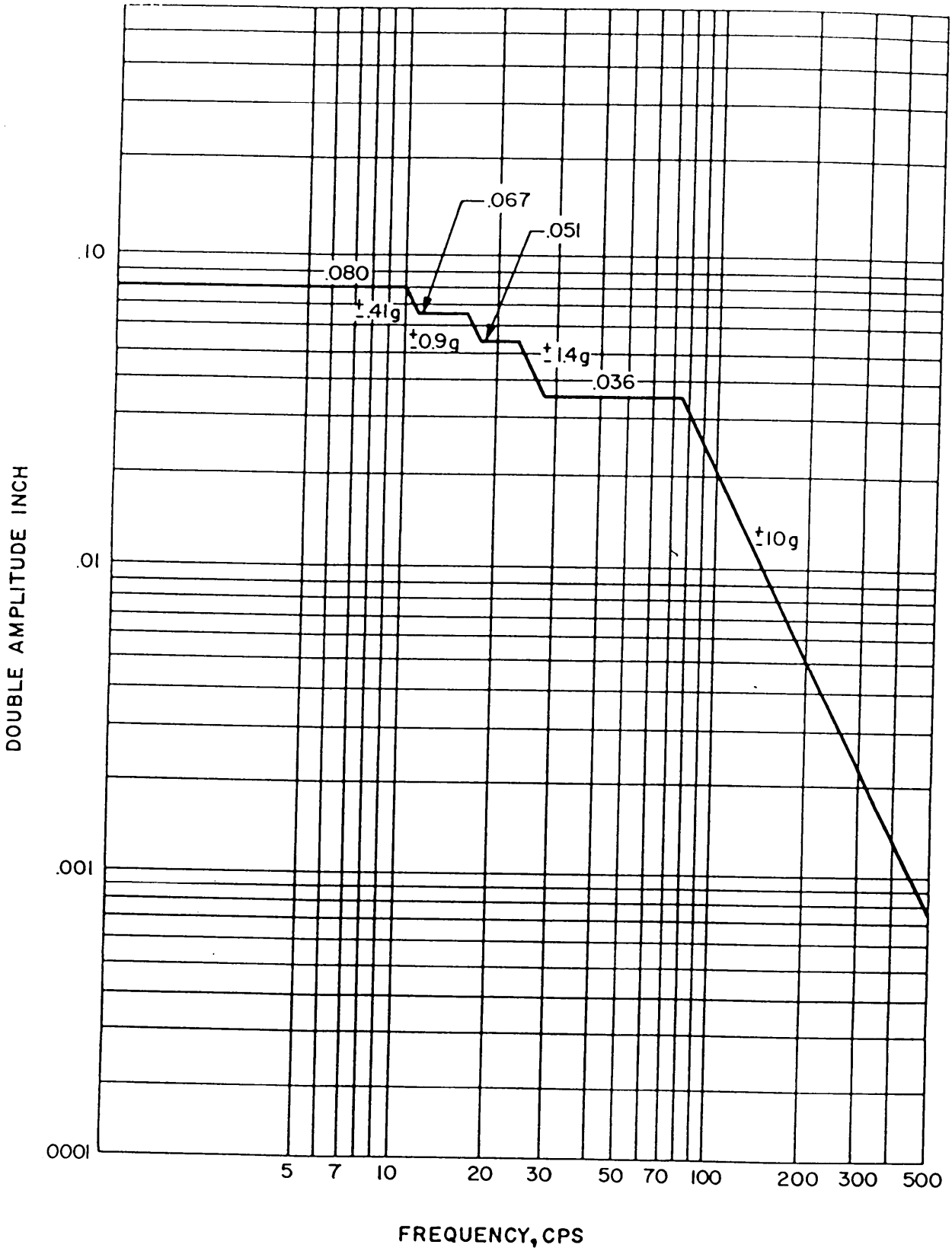
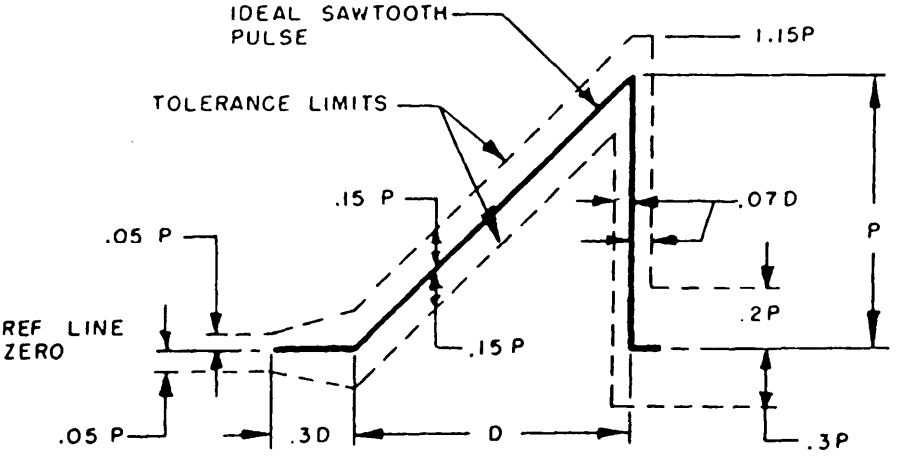


FIGURE 6. Range curve for converter vibration.

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Procedure	Inspection	Peak value (P) g's		Nominal duration (D) ms	
I	Basic design	15		11	
II	Crash safety	30		11	

NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which maybe obtained by integration of the acceleration pulse) shall be within the limits of Vi ±0.1 Vi, where Vi is the velocity-change associated with the ideal pulse which equals 0.5 DP. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1 D after the pulse.

FIGURE 7. Terminal-peak sawtooth shock pulse configuration and its tolerance limits.

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Navy - AS

Preparing activity:
Air Force - 71

Agent:
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